

International Marketing Analytics

Logistic regression across countries

WIRTSCHAFTS UNIVERSITÄT WIEN VIENNA UNIVERSITY OF ECONOMICS AND BUSINESS

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Today

Odds | Churn logit | Countries | Using the model | Python | In class

Theory, Parts 1 to 4

- Odds, odds ratios, the S-curve
- Reading the churn logit
- Comparing countries
- Predict, rank, target

Practice, Parts 5 and 6

- polars, statsmodels, plotnine
- Online quiz and its answers
- Coding exercise: churn in four countries
- The same tasks with Posit Assistant

Question of the day: who will cancel, and does it differ by country?

Our example: a mobile operator in four countries

The problem: a mobile operator in four countries is losing customers.

The question: who will cancel in the next 12 months, and what can we do?

4,000 customers in **Germany**	**1,500** customers in **Austria**	**3,000** customers in **Spain**	**2,500** customers in **Poland**

Churn = 1: the customer cancelled within the next 12 months; 0: the customer stayed.

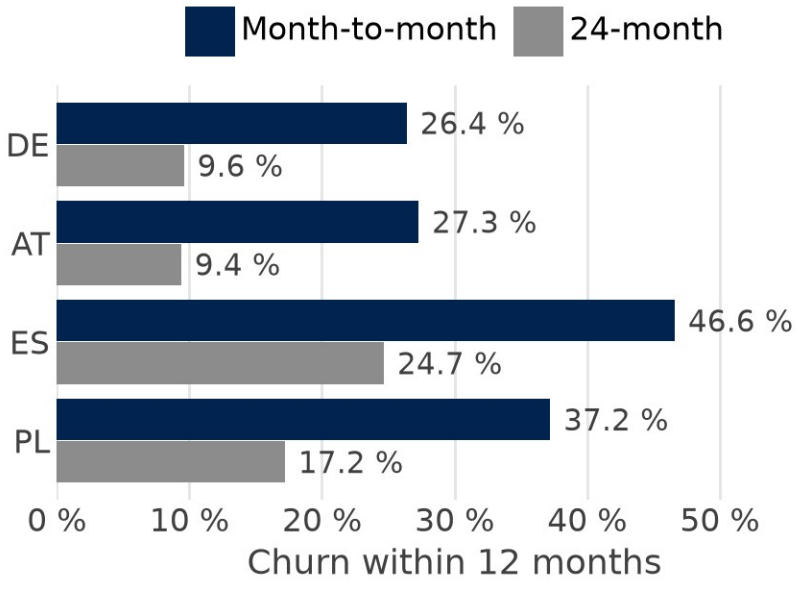
Simulated data (telco_churn.csv), built for this course

The data: one row per customer

Column	Meaning	Customer 47
customer_id	customer number	47
country	DE, AT, ES or PL	DE
age	age in years	20
tenure_months	months with the operator	10
monthly	1 = month-to-month contract, 0 = 24-month	1
monthly_fee	fee in EUR per month	27.22
price_increase	1 = price rise in the last six months	0
complaints	complaint calls in the last six months	1
satisfaction	survey score, 1 to 5	2
retention_offer	1 = got the retention offer (random pilot)	0
churn	1 = cancelled within the next 12 months	1

File: `data/churn/telco_churn.csv` simulated, 11,000 customers

First look: churn by country and contract



1 in 4 **month-to-month** customers in Germany churn (26.4 %)

1 in 10 **24-month** customers in Germany churn (9.6 %)

The contract matters everywhere – how much? That needs a model for a yes/no outcome.

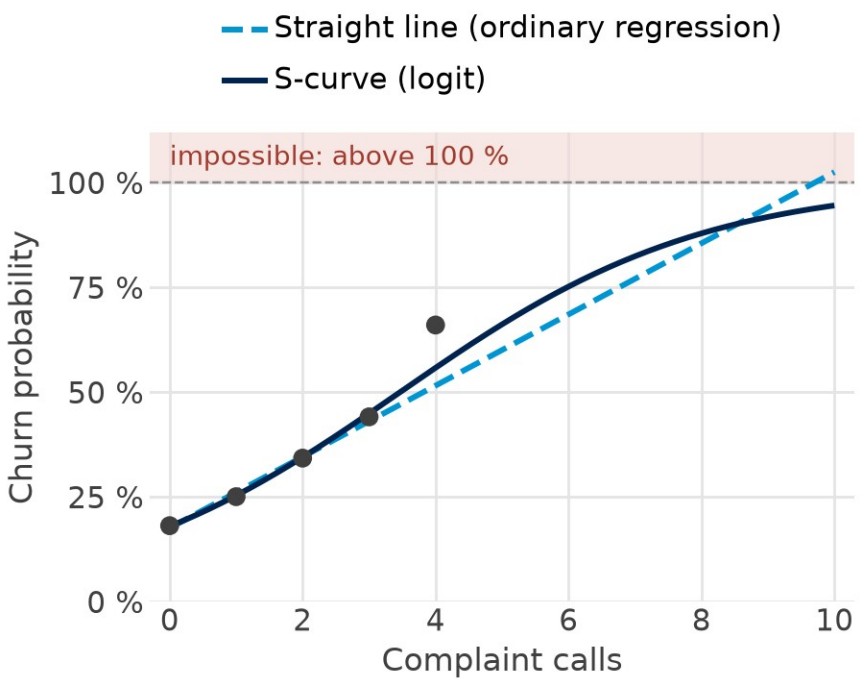
Simulated data (telco_churn.csv)

From churn rates to odds



Part 1 of 6 · odds, odds ratios, the S-curve

Why not ordinary regression for churn?



Straight line: easy to read, but no limits: 103 % churn at 10 complaint calls.

S-curve (logit): always between 0 and 100 %; at 10 calls 95 %.

The price of the S-curve: a coefficient no longer says directly how much the probability changes.

Simulated data (telco_churn.csv); dots: churn rate at 0 to 4 calls

Worked example: 100 + 100 German customers

German customers	Churners	Customers	Probability p	Odds = p / (1 − p)
Month-to-month	25	100	0.25	0.25 / 0.75 = 0.33
24-month	10	100	0.10	0.10 / 0.90 = 0.11

Formula

```
p    = churners / customers
odds = p / (1 − p)
     = churners / stayers
```

Odds 0.33 = 1/3: one churner per three stayers

Odds 0.11 = 1/9: one churner per nine stayers

Odds = churners per stayer: 0 means nobody churns, 1 means 50:50.

Invented round numbers, close to the simulated data (Germany: 26 % vs 10 %)

Odds ratio and log-odds: the same example

3.0

Odds ratio = 0.33 / 0.11
a month-to-month contract triples the odds of churn

1.10

Log-odds = ln odds: ln(1/3) − ln(1/9) = −1.10 − (−2.20)
= ln 3: the logit coefficient of month-to-month

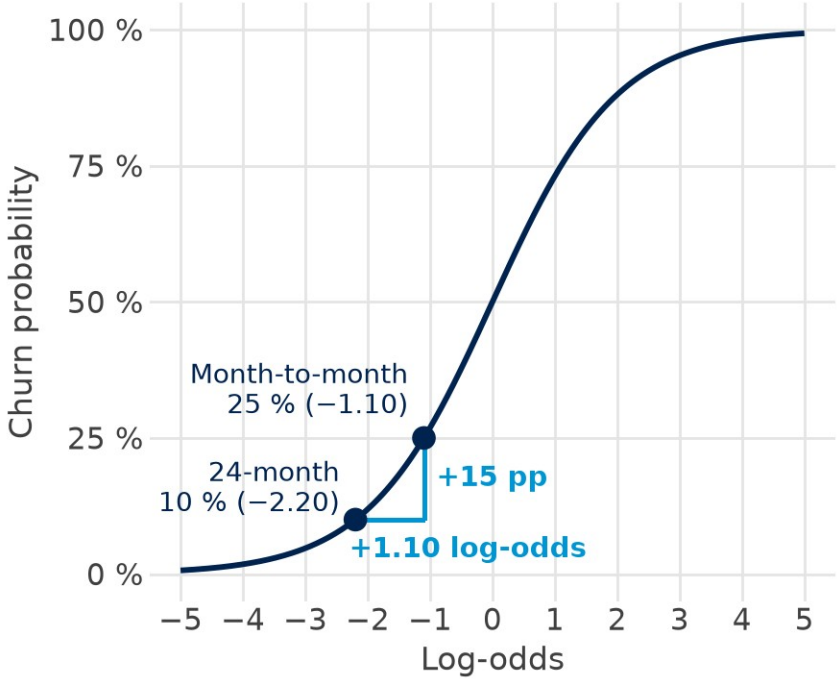
0.25

Back to probability: p = odds / (1 + odds)
0.33 / 1.33 = 0.25

Not “3 times as likely”: churn rises from 10 % to 25 %: 2.5 times, +15 percentage points (pp).

Invented round numbers, close to the simulated data (slide 8)

One model, three scales



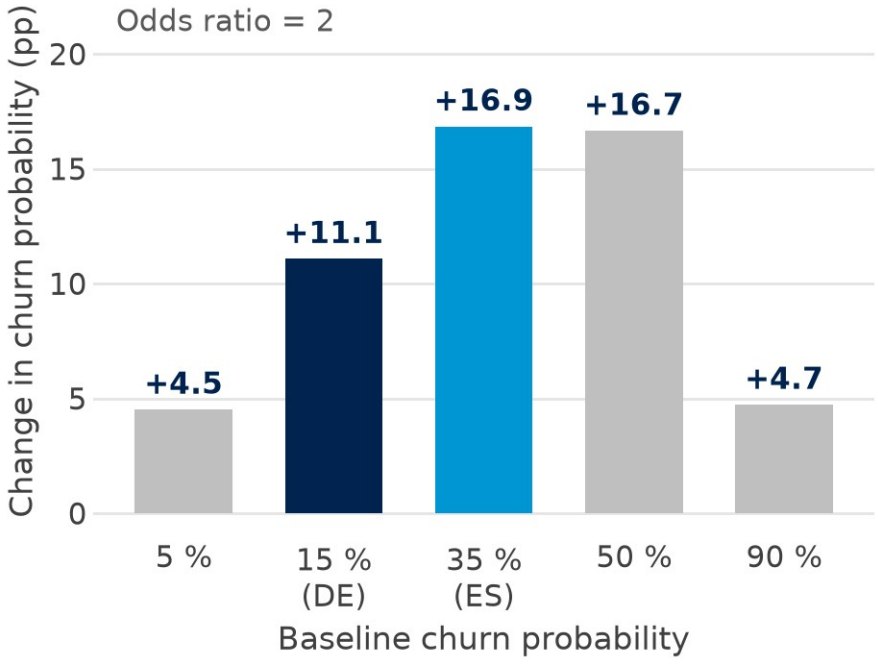
Log-odds: −2.20 → −1.10, +1.10. Effects add up; the output table shows these.

Odds: 0.11 → 0.33, × 3.0. Effects multiply: the odds ratio.

Probability: 10 % → 25 %, +15 pp. What a manager wants; the change depends on the start.

Worked example of slide 8; curve: p = 1 / (1 + e^(−log-odds))

An odds ratio of 2 is not “twice as likely”



× 1.9 **5 % → 9.5 %**
nearly twice as likely

× 1.5 **35 % → 51.9 %**
1.5 times as likely, +16.9 pp

“Twice as likely” only works for rare outcomes, below about 10 %. Churn of 15 to 35 % is not rare.

Own calculation: $p_1 = OR \cdot p_0 / (1 - p_0 + OR \cdot p_0)$, with p_0 the starting churn rate

Your turn: Spanish customers, with a calculator

Spanish customers	Churners	Customers	p	Odds	Log-odds
Month-to-month	45	100	?	?	?
24-month	25	100	?	?	?

1 p and odds of both groups

2 the odds ratio, and the difference in log-odds

3 compare with Germany (odds ratio 3.0, +15 pp): odds ratio and gap in percentage points

Pairs, 5 minutes: where does the contract matter more, Germany or Spain?

Invented round numbers, close to the simulated data (Spain: 47 % vs 25 %)

The churn logit

Part 2 of 6 · reading the output, three numbers to report

The churn logit: reading the table

	Term	Coefficient	Odds ratio
1	**Month-to-month**	0.962	2.62
	Price rise	0.602	1.83
2	**Complaint calls (each)**	0.452	1.57
	ln tenure (months)	−0.240	0.79
	Austria (vs Germany)	−0.010	0.99
3	**Spain (vs Germany)**	1.057	2.88
	Poland (vs Germany)	0.618	1.86

1. **Contract, OR 2.62:** month-to-month multiplies the odds of churn by 2.6

2. **Complaints, OR 1.57:** each call raises the odds by 57 %

3. **Spain, OR 2.88:** higher churn odds than Germany at equal drivers

Above 1 raises the odds, below 1 lowers them; percentage points come next.

Simulated data (telco_churn.csv); one logit for all countries, Germany = reference

Three numbers to report

Odds ratio

Multiplies the odds of churn

Use: inside one model

2.62

month-to-month

Marginal effect

Change in churn probability, in pp

Use: compare countries

+15.6 pp

month-to-month, on average

Prediction

Churn probability of one profile

Use: talk to managers

DE 38.1 %

95 % CI 34.1–42.2 %

ES 58.0 %

95 % CI 54.0–62.0 %

Report percentage points and probabilities; keep odds ratios in the table.

Simulated data (telco_churn.csv); prediction: month-to-month, price rise, one complaint, 12 months

Comparing countries

Part 3 of 6 · coefficients, probabilities, survey scores

The puzzle: where does the contract matter more?

One logit per country:
month-to-month coefficient

Germany 1.19
Spain 0.81

Does the contract matter more in Germany?
Show of hands: yes or no?

What else differs?
Churn: 15.4 % in Germany,
34.7 % in Spain

What did slide 12 show?
Odds ratios and pp can rank countries differently

Simulated data (telco_churn.csv); one logit per country, four drivers

Same effect, different coefficients

Country	True effect	Noise scale	Coefficient
Germany	1.3	1.0	1.19
Austria	1.3	1.0	1.02
Spain	1.3	1.5	0.81
Poland	1.3	1.2	0.90

Coefficients from one logit per country

Rule of thumb

```
coefficient ≈
  true effect / noise

Germany  1.3 / 1.0 = 1.30
Spain    1.3 / 1.5 ≈ 0.87
```

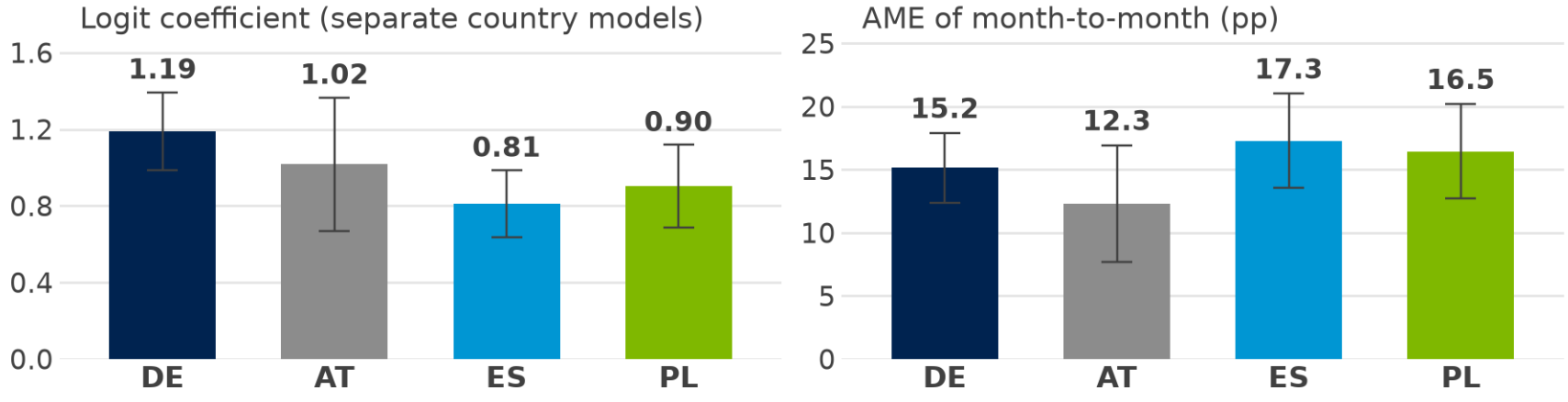
Noise scale: how much churn depends on things the model does not see

Naive test: 1.19 vs 0.81 seems a real gap ($z = 2.8$)

Smaller coefficient, same effect: more noise in Spain shrinks the coefficient.

Simulated data (telco_churn.csv); true effect and noise set in the simulation

On the probability scale the gap disappears



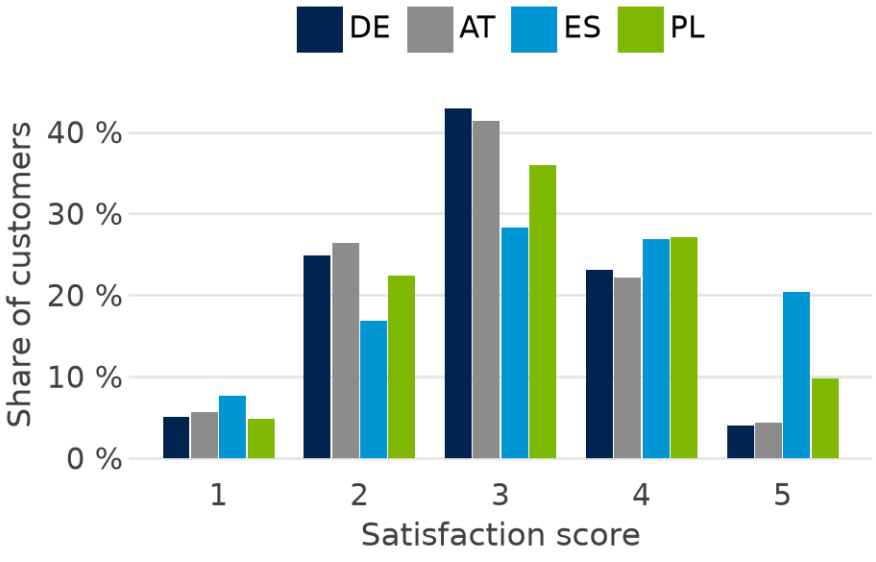
Lines: 95 % CIs. Gap Spain − Germany 2.1 pp: within sampling error

Noise: more of it shrinks Spain's coefficient (1.5 vs 1.0)

Baseline: at 34.7 % churn the S-curve is steeper than at 15.4 %

Simulated data (telco_churn.csv); one logit per country

Survey scores differ by country too



20.3 % **give a 5 in Spain** (Germany 4.0 %)

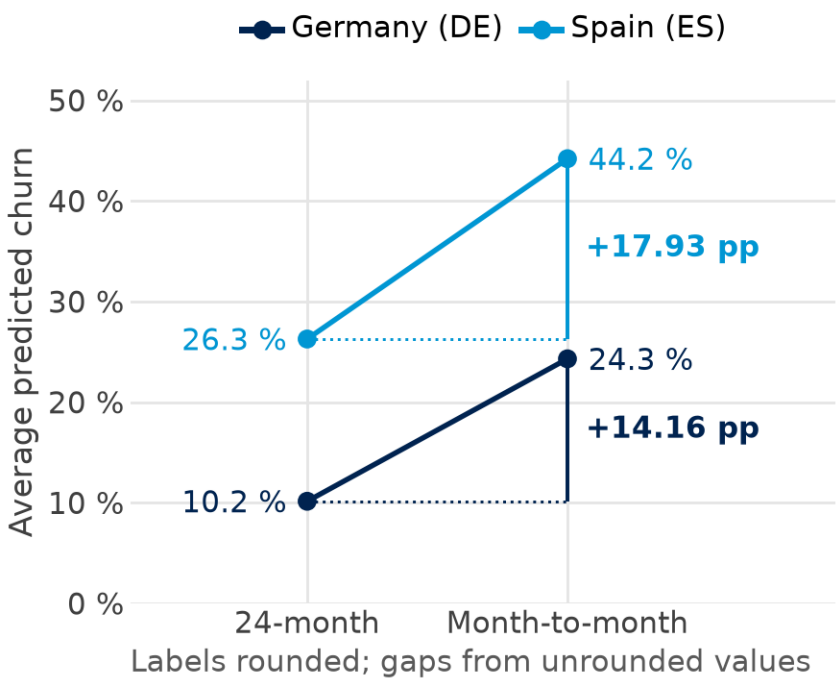
3.35 **mean score in Spain** (Germany 2.96)

Equal **true satisfaction** in every country (simulation)

Same satisfaction, different scores: standardise within each country before you pool.

Simulated data (telco_churn.csv); answer styles set in the simulation

Interactions: read them in percentage points



−0.245 **Product term** in log-odds: “weaker in Spain”

+3.8 pp **In pp:** 17.93 (Spain) minus 14.16 (Germany): “stronger”

Read interactions in pp. Here the 95 % CI (−0.7 to +7.2 pp) includes zero: no clear difference.

Simulated data (telco_churn.csv); one logit with contract × country

Using a churn model

Part 4 of 6 · predict, rank, target

Who will churn? The threshold and the confusion matrix

Fit on 70 % → Test on 30 % → **Pick a threshold** → Call the flagged

Threshold 0.5	Churned	Stayed
Flagged	175	106
Not flagged	619	2,384

77.9 % **accuracy;** “nobody churns” scores 75.8 %

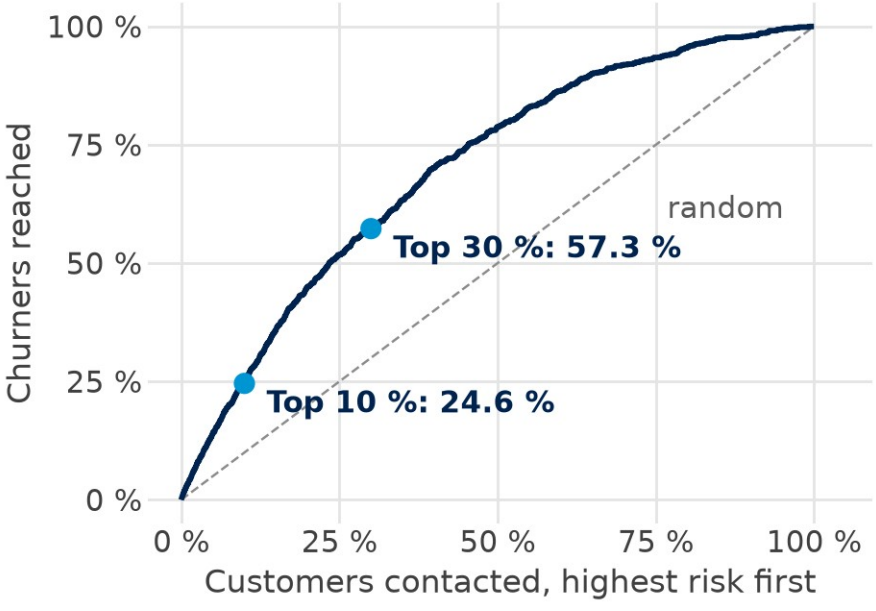
22 % **of churners caught** at 0.5; 56 % at 0.3

Flagged: predicted churn probability ≥ 0.5

Accuracy misleads when most customers stay: count the churners you catch; lower the threshold if missing one costs more than a wasted offer.

Simulated data (telco_churn.csv), 3,284 test customers

Ranking customers: the top-decile lift



2.46 — **top-decile lift:** 59.5 % churn in the top 10 %, 24.2 % overall

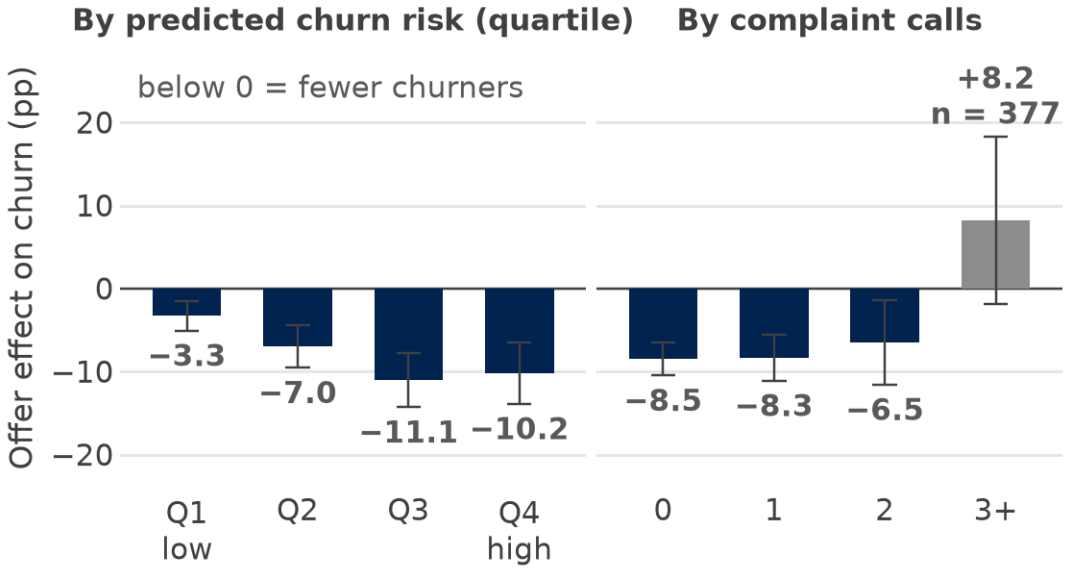
24.6 % — **of all churners** are in the top 10 %

57.3 % — **of all churners** are in the top 30 %

Fixed budget? Call from the top of the ranked list.

Sources: Neslin et al. (2006); simulated data (telco_churn.csv), test customers

Whom to call? Risk is not uplift



Pilot: a random half got a retention offer; uplift = churn with minus without it

Highest risk ≠ largest effect: Q4 −10.2 pp, Q3 −11.1 pp

No response from 3 complaint calls on (377 customers; the interval includes 0)

Call the customers the offer moves, not the riskiest: it takes a random control group.

Sources: Ascarza (2018); simulated data (telco_churn.csv), offer pilot

The analyses in Python



Part 5 of 6 · polars, statsmodels, plotnine

The packages and functions

Task	Package	Function
Load and look	polars	pl.read_csv(); group_by().agg()
Fit the logit	statsmodels	smf.logit(formula, data).fit()
Odds ratios	numpy	np.exp(m.params)
Marginal effects (pp)	statsmodels	m.get_margeff(dummy=True)
Predictions with CI	statsmodels	m.get_prediction(profiles).summary_frame()
Test split and lift	numpy	np.random.default_rng(1); np.argsort(-p)
Plots	plotnine	ggplot, geom_line, geom_ribbon

Step 1: load and look

Python: polars

```
import polars as pl

churn = pl.read_csv("../../data/churn/telco_churn.csv")
df = churn.to_pandas()

overview = (churn
    .group_by("country", maintain_order=True)
    .agg(pl.col("churn").mean().alias("churn_rate"),
         pl.col("churn").filter(pl.col("monthly") == 1)
         .mean().alias("churn_month_to_month"),
         pl.col("churn").filter(pl.col("monthly") == 0)
         .mean().alias("churn_24_month")))
overview
```

	All	Month-to-month	24-month
DE	15.4 %	26.4 %	9.6 %
AT	14.6 %	27.3 %	9.4 %
ES	34.7 %	46.6 %	24.7 %
PL	28.5 %	37.2 %	17.2 %

Churn rate by country and contract

Look first: month-to-month customers churn two to three times as often, in every country.

Step 2: fit the logit and the marginal effects

Python: statsmodels

```
import numpy as np
import statsmodels.formula.api as smf

m = smf.logit(
    "churn ~ monthly + price_increase"
    " + complaints + np.log(tenure_months)"
    " + C(country, Treatment('DE'))",
    data=df).fit()

m.get_margeff(dummy=True).summary()
ame(m, df, "monthly")    # by country
m.get_prediction(profiles).summary_frame()
```

Line	Month-to-month
get_margeff	+15.6 pp, all customers
ame	DE 12.4 pp, ES 20.1 pp
get_prediction	DE 35.1 %, ES 60.9 %

Profile: price rise, 1 complaint call, 12 months

One slope for all countries: it overstates the gap; compare countries with country models.

Step 3: evaluate the model

Python: statsmodels and numpy

```
# FORMULA: the churn formula of Step 2 (qmd Step 0)
rng = np.random.default_rng(1)
test = rng.random(len(df)) < 0.3
fit = smf.logit(FORMULA + " + satisfaction + C(country)",
                data=df[~test]).fit()
p = fit.predict(df[test]).to_numpy()
y = df.loc[test, "churn"].to_numpy()

top = np.argsort(-p)[:round(0.1 * len(y))]
lift = y[top].mean() / y.mean()          # 2.46
```

Result	Value
Test customers	3,284
Churn, all	24.2 %
Churn, top 10 %	59.5 %
Top-decile lift	2.46



Judge on new customers: fit on 70 %, test on the other 30 %.

In-class exercise

Part 6 of 6 · quiz, coding exercise, Posit Assistant

Online quiz: logit and churn in five questions

1 In Austria, 20 of 100 month-to-month customers churn. The odds of churn are …
A 0.20 B 0.25 C 4.0

2 An odds ratio of 2 at a baseline churn rate of 20 %. The new churn probability is …
A 33 % B 40 % C 22 %

3 Month-to-month coefficient: 1.19 in Germany, 0.81 in Spain. So …
A bigger effect in Germany B not clear yet: compare in pp C bigger effect in Spain

4 Product term month-to-month × Spain: −0.245. In pp, the effect in Spain …
A is smaller than in Germany B can still be larger than in Germany C is zero

5 Who should get the retention offer?
A the highest predicted risk B those whose churn it lowers most C most complaint calls

3 minutes alone, then compare with your neighbour.

Answers: in class

Your turn: churn across four countries, in Python

Pairs, with packages: 35 minutes	Plenary, answers: 5 minutes	Posit Assistant: 10 minutes

1. **Fit:** the pooled churn logit with country dummies; the month-to-month AME per country (Step 2)
2. **Compare:** the AMEs with the separate country coefficients; two memo sentences (Part 3)
3. **Evaluate:** confusion matrix at 0.5 and top-decile lift on the 30 % test set (Step 3)
4. **Target:** the offer effect by complaint calls; who should get the offer? (Step 4)

Deliverable: one slide per pair with a plot and two sentences for the retention manager.

File: `sessions/01-foundations/churn_exercise.qmd` (code to reuse: logit_churn.qmd)

Then: the same tasks with Posit Assistant

How: code first, then open Posit Assistant, type @logit_churn.qmd, one task per prompt

Prompt for task 1

```
@logit_churn.qmd Task 1. Fit the pooled
churn logit with Treatment('DE') and give
the month-to-month AME per country
in percentage points.
```

Prompt for task 2

```
Task 2. Put these AMEs next to the
separate country coefficients. Why can
the two rank Germany and Spain
differently?
```

Prompt for task 3

```
Task 3. On the 30 % test set, compute
the confusion matrix at 0.5 and the
top-decile lift. Explain both in one
sentence each for a manager.
```

Prompt for task 4

```
Task 4. Estimate the offer effect by
complaint calls from the pilot. Who
should get the offer, and why not
the riskiest customers?
```



Check every number against your own code: you own the result.

Open churn data for your project

IBM Telco Churn

7,043 customers, USA (fictional)

Churn: Left in the last month

Use: Learn the mechanics

Licence: Credit IBM

Bank churn

Churn_Modelling: 10,000 customers in France, Germany, Spain

Churn: Closed the account

Use: A country effect

Licence: Unclear: link only

Online Retail II

1.07 m transactions, 43 countries

Churn: You define it

Use: Real data, own label

Licence: CC BY 4.0

The licence decides: commit only data you may share; link the rest.

Guide: `sessions/01-foundations/churn_readings_and_data.md` (links, polars code, traps)

Sources: dataset checks, Oct 2026 (churn_readings_and_data.md)

Key takeaways

1. **Odds are not probabilities:** odds = churners per stayer; OR 3 is not “3 times as likely”
2. **Report three numbers:** odds ratio, AME in percentage points, predicted probability with CI
3. **Compare countries in pp:** not in coefficients, which carry each country's noise
4. **Interactions and survey scores:** read interactions in pp; check scores before pooling
5. **Rank by risk (lift), target by uplift:** the offer goes to those it changes

Next: run sessions/01-foundations/logit_churn.qmd; it reproduces the model results.

References

Logit methods

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Ascarza, E. et al. (2018). In pursuit of enhanced customer retention management: review, key issues, and future directions. Customer Needs and Solutions, 5(1–2), 65–81.

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Verbeke, W. et al. (2012). New insights into churn prediction in the telecommunication sector: a profit driven data mining approach. EJOR, 218(1), 211–229.

Ascarza, E. (2018). Retention futility: targeting high-risk customers might be ineffective. JMR, 55(1), 80–98.

Full list with DOIs: `sessions/01-foundations/churn_readings_and_data.md`

Thank you. Questions and discussion

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